

WILLIAM YANG

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EDUCATION

University of Waterloo

Honours Bachelor of Software Engineering, Co-op

Sep 2025 - Present

GPA: 88/100

SKILLS

Languages: Python, TypeScript, C++, SQL, Bash

Frameworks: FastAPI, Django, Express, Hono, DrizzleORM, Zod, Next.js, React, React Native

Infrastructure: Docker, AWS, Cloudflare, Vercel, PostgreSQL, MySQL, Elasticsearch, CI/CD, Linux, Git

EXPERIENCE

Nokia

Software Engineer Intern

May 2026 - Aug 2026

Ottawa, Canada

- Migrated legacy **Network Services Platform** regression tooling to **Django REST Framework** and **React/TanStack**, building **30+** endpoints and **16+** pages that launch, track, and gate **3,000+** weekly regression runs
- Shipped an **agentic AI** service diagnosing **Kubernetes** provisioning failures behind **700+** weekly runs, probing over **SSH** before control-host teardown
- Backfilled **MySQL** test failure history into a new **Elasticsearch** index using deterministic hash IDs for idempotent re-imports, along with restore-verified daily snapshots across 6 production indices (**192GB**)
- Cut analytics API latency from **9s** to **0.77s** on a **187M-document Elasticsearch** index via cache-stable time windows and bounded aggregations

WATonomous

Software Developer (WATcloud)

Sep 2025 - Present

Waterloo, ON

- Hardened **SLURM** job-completion notification service with persistent **SQLite** state on Ceph-backed **Kubernetes** PersistentVolumeClaim, eliminating **~100** monthly dropped notifications across pod restarts
- Engineered Go CLI tooling for the cluster-wide WATcloud CLI with Python parser of live **SLURM** scheduler state, delivering real-time CPU/memory/GPU capacity to **1,000+** users across **26 nodes**
- Migrated the user-directory identity model to durable GitHub global IDs via an idempotent API backfill across **350+** accounts, eliminating account-management failures from upstream username changes

PROJECTS

Canadian Computing Competition Solutions Repository

Next.js, Hono, Cloudflare Workers, Supabase, Drizzle ORM, R2 Object Storage

Sep 2024 - Present



- Modernized a static solutions archive into a comprehensive platform serving **300+** solutions to **1,000+** monthly users
- Built a **Cloudflare Workers + Hono** edge API powering an interactive forum and serving **9,000+** test-case files from private **R2** object storage via presigned URLs
- Architected a **PostgreSQL** backend with **Drizzle ORM**, **30+ Zod**-validated endpoints, and CI-gated migrations, enforcing row-level security through per-request **JWT** claims
- Cached requests at the **Workers** edge and gated writes with **Cloudflare Turnstile** CAPTCHA and per-IP rate limiting

Blockd - macOS Website Blocker Daemon

Rust, DNS, launchd, ratatui

Jun 2026 - Present

- Built a **DNS**-level website blocker in **Rust** that intercepts system queries through a hand-parsed UDP resolver, returning **NXDOMAIN** for domains on a user-defined blocklist
- Hardened the blocker against removal with a root-owned **launchd** daemon that reasserts DNS and **/etc/hosts** configuration on any tampering
- Built a **ratatui** terminal UI for blocklist management and gated every unblock behind a **TOTP** code, requiring a second party's approval to re-enable blocked sites

AWARDS

Hack the North 2025 Finalist - Top 12 of 1000+ Hackers

Sep 2025

- Built Maatchaa, a **RAG**-powered sponsorship platform matching Shopify merchants to YouTube creators